EE3032 Introduction to VLSI Design—Final

Note: Total number of points is 150, but the number of points that you can get is smaller than 100.

1. (40 points) Answer the following questions.
   (a) List two approaches which can be used to alleviate the occurrence probability of latchup. (4 points)
   (b) Explain the following definitions: (1) physical design, (2) instantiate a cell, and (3) instance. (6 points)
   (c) List two requirements which must be met when you draw the layout of a cell in the cell library. (4 points)
   (d) What is soft error? (2 points)
   (e) Why is the soft error rate of a circuit increased with the increasing of operation frequency of the circuit and the scaling down of technology process? Explain the cause in terms of the following three effects: logical masking effect, electrical masking effect, and latch window effect. (6 points)
   (f) List two methods which can be used to reduce the power bounce and ground bounce occurring in the IO pads. (4 points)
   (g) List two major problems of the dynamic logic. (4 points)
   (h) List two limitations of the CMOS domino logic. (4 points)
   (i) What are the core-limited design and the pad-limited design? (4 points)
   (j) A pseudo-NMOS logic is called a ratioed logic. Why? (2 points)

2. (10 points) Sketch the Boolean function $f = \overline{AB(C+D)}$ using each of the following logic design techniques: (1) Dynamic logic (5 points) and (2) Fully differential logic (5 points).

3. (15 points) Figure 1 shows a domino logic gate. Assume that $0.18\mu m$ technology is used to design the domino logic gate and $\mu_n = 3\mu_p$. Also, a $2\mu m/0.18\mu m$ PMOS/NMOS has an input capacitance C, i.e., $C_g = C$. Complete the following questions.
   (a) Assume that the size of P1 is $4\mu m/0.18\mu m$ and the channel length of all the transistors is equal to $0.18\mu m$. Determine the sizes of the NMOS transistors N1, N2, N3, N4, N5, and N6 such that the rise time and the fall time of the node X are equal, i.e., $t_r = t_f$. (5 points)
   (b) If your boss ask you to design the dynamic gate such that the switching power of the clock (CLK) is $2fCV_{dd}^2$, where f and V_{dd} denote the operation frequency and the power supply of the logic gate. Determine the sizes of the NMOS transistors N1, N2, N3, N4, N5, N6, and P1 such that the rise time and the fall time of the node X are equal, i.e., $t_r = t_f$. (10 points)

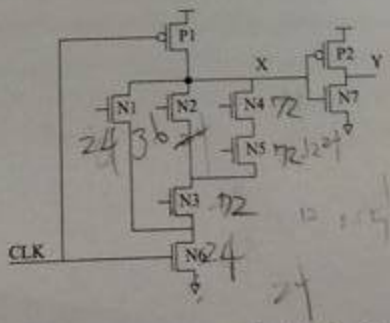


Figure 1: A dynamic logic circuit

4. (15 points) Figure 2 shows a combinational circuit designed with NP-domino logic. Assume that all the NMOS transistors have the same β_n and all the PMOS transistors have the same β_p, and $\beta_n = \beta_p$. Also, $\mu_n = 2\mu_p$ and the gate capacitance of the PMOS and NMOS transistors are $2C$ and C, respectively; and the turn-on equivalent resistance of the NMOS transistor or the PMOS transistor is R. Complete the following questions.

(a) Assume that the rise time t_r or fall time t_f of a logic gate can be estimated by the product of the output capacitance of the logic gate and the equivalent resistance of the output charge path or the output discharge path, respectively. Also, the delay t_d of a logic gate can be calculated by $t_d = (t_r + t_f)/2$ and the drain capacitance of the transistors can be neglected. Calculate the contamination delay t_{cd} and the propagation delay t_{pd} of the combinational circuit in terms of R and C. (10 points)

(b) Assume that the V_{dd} of the combinational circuit is $1.8V$ and the threshold voltage of the PMOS transistor is $|V_{tp}| = 0.5V$. Also, the capacitance of the nodes x, y, and z are 0.4pf, 0.8pf, and 1.6pf, respectively. Can the combinational circuit be operated correctly? Explain your answer. (5 points)

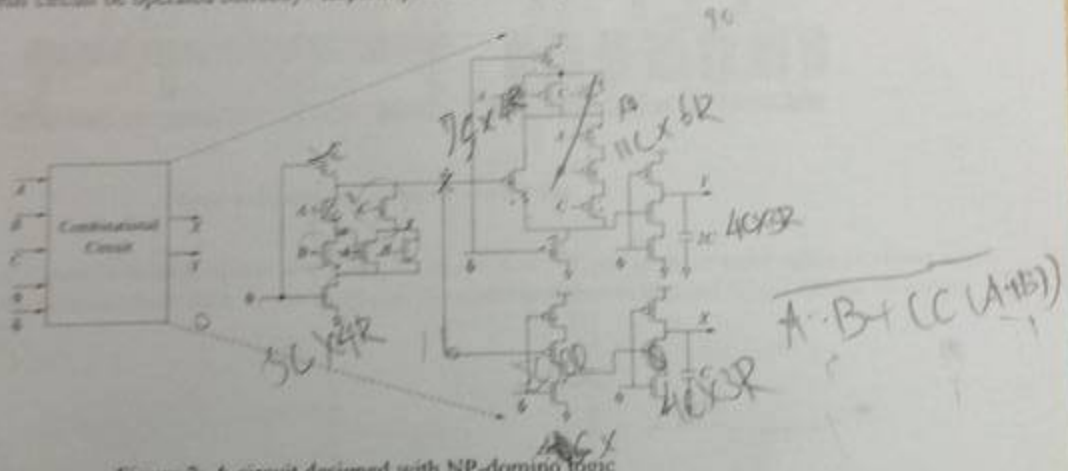


Figure 2: A circuit designed with NP-domino logic

5. (25 points) Figure 3 shows a sequential circuit. Assume that the clock-to-Q propagation delay (t_{pcq}), the clock-to-Q contamination delay (t_{ccq}), the setup time (t_{setup}), and the hold time (t_{hold}) of the registers are $0.3ns$, $0.1ns$, $0.2ns$, and $0.4ns$, respectively. Also, the delay of the NOR gate, the XOR gate, the XNOR gate, and the inverter is $0.2ns$, $0.5ns$, $0.4ns$, and $0.2ns$, respectively. Complete the following questions.

(a) Assume that clock skew of the circuit is $0ns$. Can the circuit be operated correctly? If not, how do you modify the circuit such that the circuit can operate correctly without sacrifice the operation frequency? (5 points)

(b) If clock skew is $0ns$, determine the maximum operation frequency of the circuit. (5 points)

(c) If clock skew is $0.5ns$, determine the maximum operation frequency of the sequential circuit. (5 points)

(d) If you can insert three additional registers into the circuit to boost its operation frequency, where the registers should be inserted such that maximum improvement of operation frequency can be achieved. Draw the modified circuit and calculate the maximum delay of the modify circuit. (10 points)

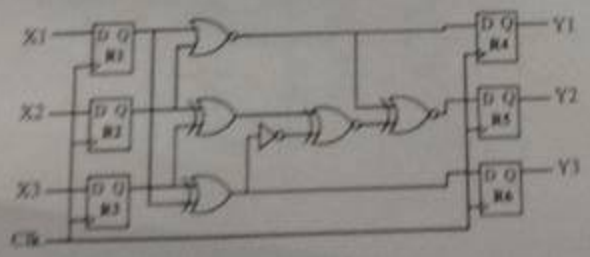


Figure 3: A sequential circuit

6. (20 points) Figure 4 shows two symbolic layouts of two circuits. Draw the transistor-level schematic diagrams of the two circuits.

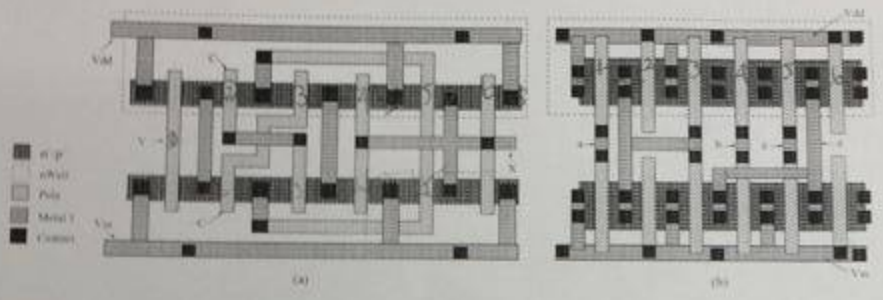


Figure 4: Symbolic layouts of two circuits

7. (10 points) Figure 5 shows a Schmitt circuit, where $V_{tn} = |V_{tp}| = 0.6V$. If you apply an input signal as shown in the left figure to the Schmitt circuit, then the circuit outputs the signal as shown in the right figure. Derive the ratios of the $\frac{\beta_{N1}}{\beta_{N2}}$ and $\frac{\beta_{P1}}{\beta_{P2}}$.

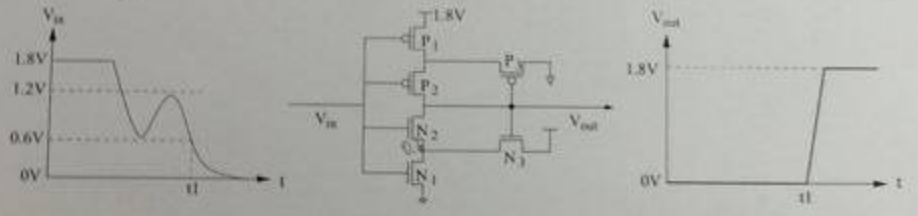


Figure 5: A CMOS Schmitt circuit

8. (15 points) Figure 6 shows a register realized by two low-cost latches. Complete the following questions.

   (a) What type of the register is? Explain your answer. (5 points)

   (b) Assume that the switching threshold voltage of the inverter Inv1 is $V_{dd}/2$ and the turn-on equivalent resistances of the transmission gate, N1, N2, P1, and P2 are R_{TS}, R_{N1}, R_{N2}, R_{P1}, and R_{P2}, respectively. List two inequalities for determining the sizes of P2 and N2 such that the register can be operated correctly. (10 points)

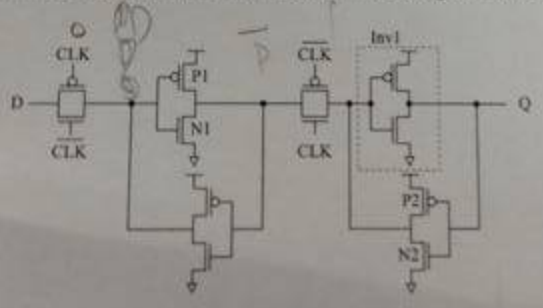


Figure 6: A register consists of two low-cost latches